Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: **Bachelor of Science in Agricultural and Biosystems Engineering**
Major in AB Machinery and Power Engineering

Primary Job Roles students are prepared for in the program	• Agricultural and Biosystems Engineer at: ○ Local Government Unit (LGU), Government Owned and Controlled Corporation (GOCC), and National Government ○ Private and Multinational Company • Technical Consultant for Agricultural and Biosystems Industries ○ o Farm Machinery Operation
Secondary Job Roles students are prepared for in the program	AB Researcher, Land Surveyor Assistant, Agricultural Extension worker, AB Structures Designer / CAD specialist, GIS specialist, Landscape Engineer, Environmental Planner, Bank Appraiser, Sales Manager (Agriculture related), Irrigation Engineer/Drainage Engineer, Agricultural Machinery Test Engineer, Designer/Manufacturer of Agricultural Machineries, Academician in ABE institution, Supervisor/Manager (Agricultural Production, and allied fields), Businessman/Entrepreneur for Agricultural and Biosystems Industries, Agrometeorology Forecaster

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program: CHED Memorandum Order (CMO) No. 94 Series of 2017
Program Description:

The BSABE program is designed to produce graduates who possess knowledge, skills, and attitudes in the application of engineering science and designs to the processes and systems involved in the sustainable production, post production, and processing of safe food, feed, fiber, timber, and other agricultural and biological materials; the efficient utilization, conservation, and management of natural and renewable resources; and development of climate change mitigation measures, in order to enhance human health in harmony with the environment. Agricultural and Biosystems (AB) consist of crops, poultry, livestock, fisheries and aquaculture resources, forestry and other plants, new and renewable energy, wastes, natural resources, and climate.

The AB Machinery and Power Engineering major will emphasize areas for specialization in Instrumentation in farm machinery, ergonomics and safety in farm operation, testing and evaluation of tractors and farm equipment, energy conservation and management in farm power and machinery.

	Program Educational Objectives	Mission-Vision and Core Value Alignment	Reference Source or CMO Provision Compliance
PEO1	Provide leadership in planning, implementing, and monitoring ABE projects and programs	Vision: A Premier University in the ASEAN Region Mission: USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management. • Mission 2: Provide quality education for the students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work. Core: Unity, Stewardship, Excellence, Professionalism	CHED Memorandum Order (CMO) No. 94 Series of 2017
PEO2	Occupy supervisory positions in private and public organizations; locally and internationally	Vision: A Premier University in the ASEAN Region Mission: USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management. • Mission 2: Provide quality education for the students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work. Core: Unity, Stewardship, Excellence, Professionalism	CHED Memorandum Order (CMO) No. 94 Series of 2017
PEO3	Own and/or manage ABE-based business enterprises	Vision: A Premier University in the ASEAN Region Mission: USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management. • Mission 2: Provide quality education for the students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work. • Mission 3: Promote entrepreneurship and industry collaboration. Core: Unity, Stewardship, Excellence, Professionalism	CHED Memorandum Order (CMO) No. 94 Series of 2017
PEO4	Pursue advance studies in ABE and emerging related fields	Vision: A Premier University in the ASEAN Region Mission: USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management. • Mission 1: Engage in high impact research, not only for knowledge's sake, but also for its practical benefits to society. Core: Unity, Stewardship, Excellence, Professionalism	CHED Memorandum Order (CMO) No. 94 Series of 2017
PEO5	Occupy responsible positions in ABE education	Vision: A Premier University in the ASEAN Region Mission: USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management. • Mission 2: Provide quality education for the students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work. Core: Unity, Stewardship, Excellence, Professionalism	CHED Memorandum Order (CMO) No. 94 Series of 2017

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Institutional Graduate Attributes:
IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

Program Attribute		Program Outcomes	Performance Indicators
IGA1	a	Apply knowledge of mathematics and science to solve complex AB engineering problems	1. Apply mathematics in designing agricultural structures, irrigation systems, and agricultural machineries 2. Apply knowledge in Physics, thermodynamics, and heat transfer in designing agricultural drying and storage facilities 3. Apply chemistry & biological sciences in designing cold storage facilities & biological wastes recovery systems
IGA1	b	Design and conduct experiments, as well as to analyze and interpret data	1. Formulate and design research proposal 2. Conduct scientific research under laboratory and field conditions 3. Analyze, interpret, and write research results
IGA1, IGA4	c	Design a system, component, or process to meet desired needs within realistic constraints, in accordance with standards	1. Design agricultural structures and electrical systems using computer-aided design software 2. Design irrigation systems for cash, annual, and plantation crops 3. Design, fabricate, and evaluate performance of agricultural machineries and equipment
IGA4, IGA2	d	Function in multidisciplinary teams	1. Demonstrate to work in a team as team member and/or as team leader 2. Deliver assigned tasks promptly and comprehensively 3. Initiate conduct of group-based laboratory experiments in multi-disciplinary engineering subjects
IGA1, IGA4	e	Identify, formulate, and solve complex AB engineering problems	1. Identify researchable areas in at least three (3) major fields in Agricultural Engineering 2. Formulate solutions to the specified engineering problems identified 3. Produce at least one simulated model as solution to an engineering problem
IGA4	f	Understand professional and ethical responsibility	1. Identify relevant laws in the practice of the profession and the duties and responsibilities of Agricultural Engineers 2. Describe the important provisions of laws and the responsibilities of Agricultural Engineers 3. Comprehend the moral issues and ethical responsibilities of agricultural engineers as practitioner and as member of an organization
IGA5, IGA2	g	Communicate effectively complex AB engineering activities	1. Produce technical-based reaction papers and reports 2. Produce and present effective oral communication materials 3. Adapt presentation styles and contents to match specific audiences
IGA3, IGA4	h	Understand the impact of engineering solutions in a global, economic, environmental and societal context	1. Identify issues and impacts of agricultural engineering practices and activities using online and other data sources 2. Prepare feasibility study of one engineering community-based project 3. Organize technical seminars and evaluate the results
IGA6	i	Recognize the need for, and an ability to engage in life-long learning	1. Evaluate community-based engineering problem and recommend solutions 2. Prepare feasibility study of one engineering community-based project

			3. Organize technical seminars and evaluate the results
IGA3, IGA1	j	Know contemporary issues	1. Explain the role of Agricultural Engineers on social, economic, political, and environmental issues relevant to nation building, security, and stability 2. Analyze the effects of current social, economic, political, and environmental. issues to national development and stability 3. Recognize the impact of Agricultural Engineering Technologies on local, national, and international issues
IGA4	k	Use techniques, skills, and modern engineering tools necessary for engineering practice	1. Describe the functionality, operation, and application of modern agricultural engineering tools and equipment 2. Prepare an effective and systematic design procedures to enhance skills and techniques in implementing agricultural engineering practice 3. Demonstrate agricultural structure design output as a proof of development in the engineering practice.
IGA2	l	Know and understand engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments	1. Describe managerial concepts and techniques adopted by successful Technical Managers, Project Directors, & other technically oriented managers 2. Understand leadership and management skills necessary in the successful implementation of Ag Engineering projects 3. Prepare feasibility study of one engineering community-based project
IGA4	m	Understand at least one specialized field of agricultural and biosystems engineering practice	1. Exemplify the concepts, theories and engineering practices in farm power and machinery 2. Exemplify the concepts, theories and engineering practices in soil and water conservation 3. Exemplify the concepts, theories and engineering practices in crop processing, postharvest technology and renewable energy technology 4. Exemplify the concepts, theories and engineering practices in agricultural structures and environmental management

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

List courses such that first year first semester courses appear first, until the last semester of the last year of the degree program. Exclude in the list the following: (a) General Education courses (b) Elective courses. As much as possible, observe the following:

f. For every course of 3-units, do not exceed 3-4 Graduate Outcomes you will be focusing and targeting to develop in the course
g. For every course of 1-units, do not exceed 1-2 Graduate Outcomes you will be focusing and targeting to develop in the course
h. For every course of 5-units, do not exceed 4-5 Graduate Outcomes you will be focusing and targeting to develop in the course
i. Ensure that all the Graduate Outcomes have courses developing it for every year level
j. Total the number of X's horizontally per course and vertically per Graduate Outcome

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Year Level	Semester	Units	Courses in the Curriculum	a	b	c	d	e	f	g	h	i	j	k	l	m
1st	First Semester	3	Purposive Communication													
1st	First Semester	3	Mathematics in the Modern World													
1st	First Semester	1	Introduction to AB Engineering								I		I			
1st	First Semester	3	Principles of Crop Science		E		E			E				E		
1st	First Semester	1	Engineering Drawing and Plans				I			I						
1st	First Semester	4	Chemistry for Engineers	I	I			I								
1st	First Semester	3	Engineering Calculus 1	I				I								
1st	First Semester	2	Movement Enhancement													
1st	First Semester	3	National Service Training Program 1													
1st	Second Semester	3	Understanding the Self													
1st	Second Semester	3	Ethics													
1st	Second Semester	4	Physics for Engineers	E	I		I	E								
1st	Second Semester	3	Principles of Soil Science		E		E			E				E		
1st	Second Semester	3	Principles of Animal Science		E		E			E				E		
1st	Second Semester	1	Computer Aided Drafting						I	I		I		I		
1st	Second Semester	3	Engineering Calculus 2	E				E								
1st	Second Semester	2	Fitness Exercise													
1st	Second Semester	3	National Service Training Program 2													
2nd	First Semester	3	The Life and Works of Rizal													
2nd	First Semester	3	Readings in Philippine History													
2nd	First Semester	3	Differential Equations	E				E								
2nd	First Semester	3	Fluid Mechanics	E				E						E		

2nd	First Semester	3	Materials and Processes for ABE	E				E						E			
2nd	First Semester	3	Engineering Mechanics	E	E					E				E			
2nd	First Semester	3	Engineering Data Analysis	I				I			I						
2nd	First Semester	3	Engineering Economics	E					I		I		I		I		
2nd	First Semester	2	Physical Activities Towards Health & Fitness I														
2nd	Second Semester	3	Science, Technology, and Society														
2nd	Second Semester	5	Thermodynamics & Heat Transfer	E				E						E			
2nd	Second Semester	3	AB Power Engineering	D				D						D		D	
2nd	Second Semester	3	Engineering Mechanics 2	E	E					E				E			
2nd	Second Semester	3	Surveying	E	E			E						E			
2nd	Second Semester	3	Computer Applications in AB Engineering	E				E						E			
2nd	Second Semester	3	Properties of AB Materials	D						D				D		D	
2nd	Second Semester	1	Biochemistry	E	I									I			
2nd	Second Semester	2	Physical Activities Towards Health and Fitness 2														
3rd	First Semester	3	General Education Elective 1														
3rd	First Semester	3	General Education Elective 2														
3rd	First Semester	3	Arts Appreciation														
3rd	First Semester	3	Methods of Research						E	E	E		E		E		
3rd	First Semester	3	AB Machinery and Mechanization				D							D	D	D	
3rd	First Semester	3	Hydrometeorology							D			D	D		D	
3rd	First Semester	3	AB Electrification and Control Systems			D		D						D		D	
3rd	First Semester	3	Strength of Materials	E				E						E			
3rd	First Semester	1	ABE and Related Laws, Specifications, Contracts, and Professional Ethics						D		D						
3rd	Second Semester	3	General Education Elective 3														
3rd	Second Semester	3	Contemporary World														
3rd	Second Semester	3	Technopreneurship				E		E	E					E		
3rd	Second Semester	3	Machine Design for AB Production	D		D								D		D	
3rd	Second Semester	3	AB Products Processing and Storage			D					D			D		D	
3rd	Second Semester	3	AB Structures Engineering			D	D				D					D	
3rd	Second Semester	3	Irrigation and Drainage Engineering			D					D			D		D	
3rd	Second Semester	3	Structural Design (Timber, Steel & Reinforced Concrete Designs)		E	E			E					E			
3rd	Second Semester	2	Thesis* (Thesis Outlining and Outline Defense/Preliminary Conduct)		D					D							
3rd	Off Semester	3	Industry Immersion Program (240 hours)				D			D					D	D	
4th	First Semester	3	Renewable Energy for AB Applications	D		D								D		D	
4th	First Semester	3	Food Process Engineering			D					D			D		D	

4th	First Semester	3	Plant and Livestock Systems and Environmental Control Engineering			D	D			D						D
4th	First Semester	3	Land & Water Conservation Engineering			D					D			D		D
4th	First Semester	3	Design and Management of AB Processing Systems			D					D			D		D
4th	First Semester	3	Specialized Professional Course 1			D					D			D		D
4th	First Semester	3	Specialized Professional Course 2			D					D			D		D
4th	First Semester	1	Undergraduate Seminar	D	D											
4th	First Semester	2	Thesis* (Conduct of Thesis)		D					D						
4th	Second Semester	3	AB Waste Management Engineering			D					D		D			D
4th	Second Semester	3	Aquaculture Engineering			D					D			D		D
4th	Second Semester	3	Specialized Professional Course 3			D					D			D		D
4th	Second Semester	2	Correlation Course	D				D								
4th	Second Semester	2	Thesis (Thesis Writing & Final Defense)		D		D			D		D				

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	/				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	/				
Do you think that your course planning technique has gone better this time than before?	/				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	/				
Do you think your topics will be better anchored to a specific purpose/s than before?	/				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	/				
Do you think that if this course will be properly implemented, your students will be developed better?	/				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	/				

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